AI Scientist Homework

Introduction

GenBio AI's AI-Driven Digital Organism (AIDO) foundation-model stack is opening a path toward a virtual laboratory, a sandbox where researchers can design, perturb, and observe biological systems entirely *in silico*. AIDO aims to accelerate biology research by enabling cheap, fast, and easy virtual experiments powered by AI to rapidly generate, test and prioritize hypotheses. Like other computational systems, the virtual laboratory will enable researchers to go beyond the physical limitations of scientific instruments and data collection to perform counterfactual studies, generic interventions, and optimization routines on AIDO. We expect both experts and armchair biologists to benefit from a platform which democratizes access to a field which has traditionally required deep technical knowledge.

The core components of AIDO are foundation models for generation, property prediction, and translation, pretrained on various data types and scales. A key technical challenge is building and exposing these models so that they may cooperate to simulate complex multi-scale systems and form a design loop where users can zoom in, make a change, propagate its effects, and zoom out to visualize.

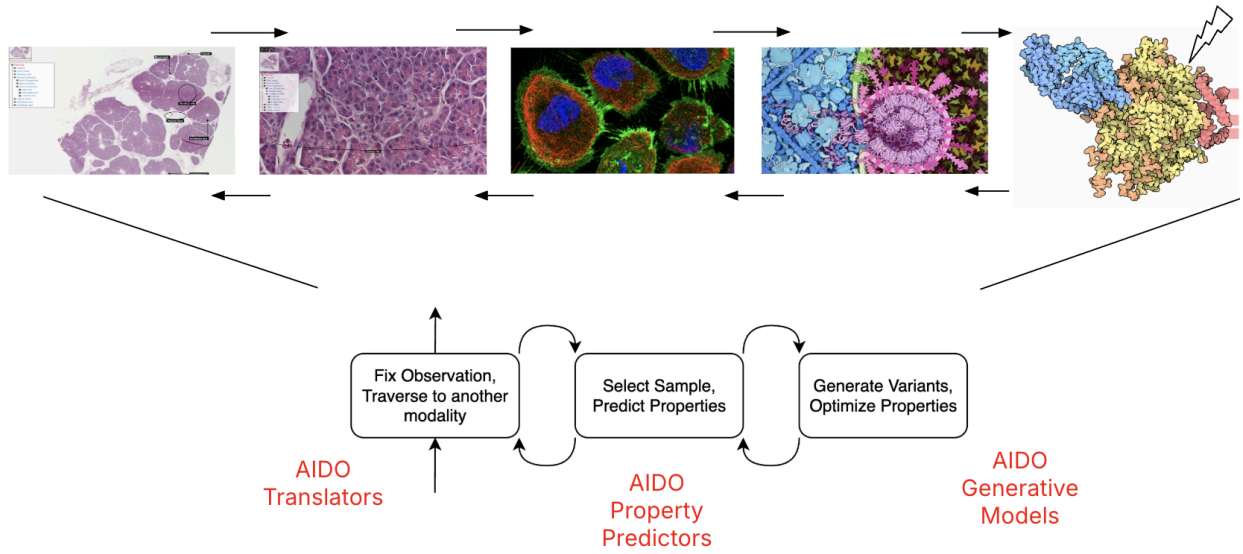

Figure 1: AIDO virtual lab system. See https://github.com/genbio-ai/AIDO for the full position paper.

Assignment

Your job as an AI scientist at GenBio AI will be to enable users to interact with AIDO through a co-scientist that guides them through experiment design and biological simulation. To demonstrate your ability, we would like you to build a language-based co-scientist that

automates the process of building new machine learning models. This co-scientist should take any dataset from the genbio-leaderboard and produce

1. A performant machine learning model
2. An accompanying codebase and model object/weights, to
    a. Reproduce training and analysis
    b. Port the model to new locations and run inference on new data
3. A final report, summarizing the process and findings

Your co-scientist should be implemented as a CLI tool, and it must provide a pleasant user experience. It must provide 2 operating modes: (i) active collaboration with the user and (ii) completely autonomous operation. Feel free to make further design choices based on how you might picture yourself using this tool in your own work.

When finished, provide (1) your codebase to us with quickstart installation instructions, so we can test your co-scientist on hidden datasets and (2) final generated codebases and reports for the 2 included tasks

- `RNA/translation_efficiency_muscle`
- `expression/cell_type_classification_segerstolpe`

Use of AI-assistants (Claude Code, Codex, Cursor, etc) is encouraged to maximize productivity, but be prepared to discuss and defend your design decisions and implementation choices in depth.